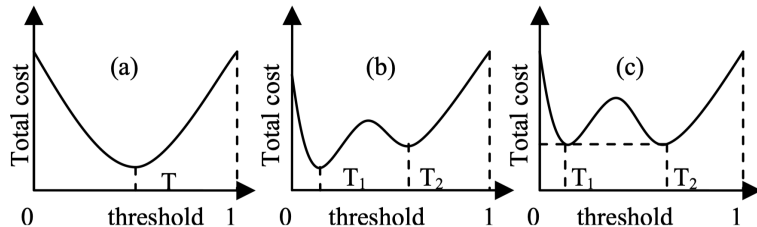
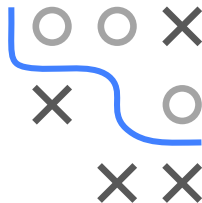


EMPIRICAL THRESHOLDING: BINARY CASE

- Theoretical threshold from MECP not always best, due to e.g. wrong model class, finite data, etc.
- Simply measure costs on data with different thresholds
- Then pick best threshold (Fig.1 in [Sheng et al. 2006](#)):

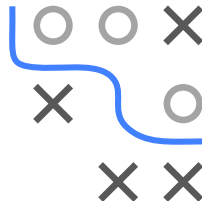


- What if two equal local minima? We prefer the one with wider span
- Do this on validation data / over cross-val to avoid overfitting!

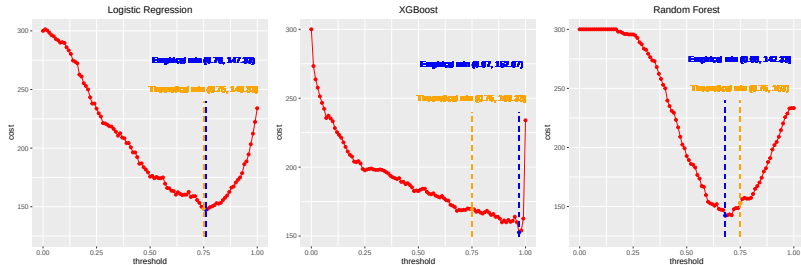
EMPIRICAL THRESHOLDING: BINARY CASE

- Example: German Credit task

	True class	
	$y = \text{good}$	$y = \text{bad}$
Pred. $\hat{y} = \text{good}$	0	3
class $\hat{y} = \text{bad}$	1	0

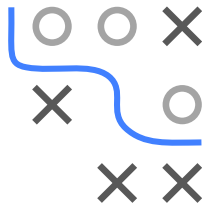


- Theoretical: $C(\text{good}, \text{bad}) / (C(\text{bad}, \text{good}) + C(\text{good}, \text{bad})) = 3/4 = c^*$
- Empirical version with 3-CV: For XGBoost, empirical minimum deviates substantially from theoretical version



EMPIRICAL THRESHOLDING: MULTICLASS

- In the standard setting, we predict class $h(\mathbf{x}) = \arg \max_k \pi_k(\mathbf{x})$.
- Let's use g thresholds c_k now
- Re-scale scores $\mathbf{s} = (\frac{\pi(\mathbf{x})_1}{c_1}, \dots, \frac{\pi(\mathbf{x})_g}{c_g})^\top$,
- Predict class $\arg \max_k \pi_k(\mathbf{x})$.
- Compute empirical costs over cross-validation
- Optimize over g (actually: $g - 1$) dimensional threshold vector $(c_1, \dots, c_g)^\top$ to produce minimal costs



METACOST: ALGORITHM

Input: $\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n$ training data, B number of bagging iterations, $\pi(\mathbf{x})$ probabilistic classifier, $\mathbf{C}$ cost matrix, empty dataset $\tilde{\mathcal{D}} = \emptyset$

Bagging: Classifier is trained on different bootstrap samples.

for $b = 1, \dots, B$ **do**
$$\mathcal{D}_b \leftarrow \text{Bootstrap version of } \mathcal{D}$$
$$\pi_b \leftarrow \text{train classifier on } \mathcal{D}_b$$

end for

Relabeling: Find classifiers for which $\mathbf{x}^{(i)}$ is OOB and compute π_b by averaging over predictions. Determine new label $\tilde{y}^{(i)}$ w.r.t. to the cost minimal class.

```

for  $i = 1, \dots, n$  do

```

$$\tilde{M} \leftarrow \bigcup_{m: \mathbf{x}^{(i)} \notin \mathcal{D}_m} \{m\}$$

end for

for $j = 1, \dots, g$ **do**
$$\pi_j(\mathbf{x}^{(i)}) \leftarrow \frac{1}{|\tilde{M}|} \sum_{m \in \tilde{M}} \pi_j(\mathbf{x}^{(i)} \mid f_m) \text{ for each } i$$

end for

for $i = 1, \dots, n$ **do**
$$\tilde{y}^{(i)} \leftarrow \arg \min_k \sum_{j=1}^g \pi_j(\mathbf{x}^{(i)}) C(k, j)$$
$$\tilde{D} \leftarrow \tilde{D} \cup \{(\mathbf{x}^{(i)}, \tilde{y}^{(i)})\}$$

end for

Cost Sensitivity: Train on relabeled data.

$$f_{meta} \leftarrow \text{train } f \text{ on } \tilde{D}$$
